

9amHealth Data Analyst Case Study Briefing

Key Business Question:

What are the key drivers of clinical improvement in body weight among 9amHealth members?

Context:

At 9amHealth, understanding what drives meaningful clinical improvements is essential to tailoring effective care and support strategies. This case study focuses on identifying which factors correlate most with body weight reduction over time, and how these insights can shape program enhancements.

You are provided with a de-identified dataset containing demographics, body weight changes, engagement activity, and progress in health education modules from a sample of members.

You will use this data to explore the impact of engagement, demographics, educational program involvement, and behavior change on weight change outcomes.

Objective:

Analyze the provided datasets and present a concise, strategic narrative that:

- Identifies key patterns and trends in weight change.
 - Identifies key patterns and trends in engagement activities.
 - Determines which engagement activities or member characteristics most strongly correlate with weight loss success.
 - Provides actionable recommendations for leadership.
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Datasets Provided:

1. **Demographics.csv**

Includes age, sex, ethnicity, subscription status, and cancellation date.

2. **BW_Detail.csv**

Contains first and last recorded body weight measurements over the defined period, weight difference, time between measurements, and total time in program.

3. **Engagement Data.csv**

Tracks member activities and engagement events over time (e.g., registration, medical questionnaire answered, subscription events).

4. **Module Progress.csv**

Records each member's completion of educational health modules, including goal setting, physical activity, emotional eating, and mindset topics.

Deliverables:

- **A 5-7 slide presentation** summarizing:
 - Analytical approach
 - Key findings
 - Visualizations (charts/tables)
 - Strategic insights and business recommendations

Candidate Responsibilities:

You are expected to:

1. Break down the key question into sub-questions and formulate hypotheses.
2. Determine which data columns are most relevant and how to structure your analysis.
3. Perform the analysis using a tool of your choice.
4. Use visuals to support your conclusions.
5. Clearly communicate implications to senior leadership.

Evaluation Criteria:

- Quality and clarity of analytical reasoning
- Insightfulness of findings
- Ability to link analysis to strategic business decisions
- Clarity and effectiveness of communication and visual presentation

Please be prepared to present your findings and answer questions from a panel that includes cross-functional stakeholders from product, clinical, and business teams.

Good luck!